ANISH BASNET

C++ Systems • Backend Engineer • Full Stack Developer

Mississauga, ON · theanishbasnet@gmail.com · linkedin.com/in/theanishbasnet · github.com/anishbasnetAB

PROFESSIONAL SUMMARY

Software engineer with a postgraduate diploma in Full Stack Software Development and hands-on experience in C++ systems programming, distributed backend architecture, and full-stack web development. Built a live distributed trading exchange simulator achieving 1,626,655 orders/sec with p50 latency of 583ns — featuring a C++20 matching engine, microservice backend, Redis event streaming, and a real-time React trading dashboard. Strong foundation in C++20, Java, Python, JavaScript, TypeScript, C#, .NET Core, OOP, data structures, REST APIs, Redis, Docker, AWS, CI/CD, and system design. Actively uses Claude Code, Codex, and Cursor daily.

SKILLS

C++ / Systems: C++20/17, STL, RAII, Smart Pointers, Move Semantics, Lock-Free Data Structures, std::atomic, Concurrency, Memory Ordering, GoogleTest, CMake, GDB, Valgrind, Linux

Backend & APIs: Java, Python, Node.js, Fastify, Express.js, ASP.NET Core 6, C#, Spring Boot (fundamentals), REST APIs, WebSocket, Microservices, Redis Pub/Sub, JWT Auth, OOP, SOLID

System Design: Redis, AWS, Docker, Docker Compose, Kubernetes (fundamentals), CI/CD, GitHub Actions, Scalable Architecture, Caching, Load Balancing, PostgreSQL, System Design Fundamentals

Full Stack: React 18, TypeScript, JavaScript, Tailwind CSS, Bootstrap, HTML5, CSS3, MongoDB, MySQL, SQL Server (fundamentals), Entity Framework (fundamentals), Vite

AI & Tools: Claude Code, OpenAI Codex, Cursor, Git, GitHub, Visual Studio, VS Code, Jira, Postman, Agile/Scrum, Unit & Integration Testing

PROJECTS

Real-Time Distributed Trading Exchange Simulator — C++20 · Node.js · Fastify · React 18 · Redis · PostgreSQL · Docker · GitHub Actions
Personal Project | May 2026 | Deployed on Railway & Vercel | Live Demo | GitHub

- Production-grade distributed trading platform: C++20 matching engine achieving 1,626,655 orders/sec · p50 583ns · p99 958ns · 11 GoogleTest unit tests · O(log n) price-time priority. Fastify microservice API gateway, JWT auth with refresh token rotation, role-based access control, pre-trade risk validation, Redis Pub/Sub event streaming, WebSocket real-time market data, atomic PostgreSQL transactions across 9 tables, Docker Compose orchestration with GitHub Actions CI.

Job Connect — Full Stack Job Portal — React · TypeScript · Node.js · Express · MongoDB
Personal Project | 2024–2025 | GitHub

- Full-stack job portal with RESTful API backend, MongoDB data layer, and React/TypeScript frontend — end-to-end delivery with Git, CI/CD, Docker, and automated testing.

Book Store Management System — ASP.NET Core 6 · C# · Entity Framework · Razor Pages · Bootstrap · MVC
Academic Project | 2023–2024 | Team Collaboration — Lambton College | GitHub

- n-Tier enterprise web application in ASP.NET Core 6 and C# with MVC architecture, Entity Framework, full CRUD operations, and responsive Bootstrap UI — built collaboratively using Visual Studio and Git.

PROFESSIONAL EXPERIENCE

NexGen | *Full Stack Developer Intern* | Jul 2023 — Dec 2023

- Built and shipped backend REST API services and frontend features across Java, JavaScript, and TypeScript — applying OOP, SOLID principles, CI/CD, and Git in an Agile team. Leveraged Claude Code and Codex daily to accelerate delivery and code quality.
- Resolved production defects through root-cause analysis; participated in code reviews, unit and integration testing, and pipeline workflows.

Technobyte | *Flutter Intern* | Dec 2021 — Mar 2022

- Shipped mobile features integrating Flutter frontends with MySQL-backed backend services across design, testing, debugging, and issue resolution.

EDUCATION

Postgraduate Diploma — Full Stack Software Development | Lambton College · 2024–2025
BSc Computing | London Metropolitan University · 2019–2022